Wheeler's guidance, so I knocked on his office door that first time with trepidation.

Professor Wheeler greeted me with a warm smile, ushered me into his office, and began immediately (as though I were an esteemed colleague, not a total novice) to discuss the mysteries of stellar implosion. The mood and content of that stirring private discussion are captured in Wheeler's writings of that era: "There have been few occasions in the history of physics when one could surmise more surely than one does now [in the study of stellar implosion] that he confronts a new phenomenon, with a mysterious nature of its own, waiting to be unravelled.... Whatever the outcome [of future studies], one feels that one has at last [in stellar implosion] a phenomenon where general relativity dramatically comes into its own, and where its fiery marriage with quantum physics will be consummated." I emerged, an hour later, a convert.

Wheeler gave inspiration to an entourage of five to ten Princeton students and postdocs—inspiration, but not detailed guidance. He presumed that we were brilliant enough to develop the details for ourselves. To each of us he suggested a first research problem—some issue that might yield a bit of new insight about stellar implosion, or black holes, or the "fiery marriage" of general relativity with quantum physics. If that first problem turned out to be too hard, he would gently nudge us in some easier direction. If it turned out easy, he would prod us to extract from it all the insight we possibly could, then write a technical article on the insight, and then move on to a more challenging problem. We soon learned to keep several problems going at once—one problem so hard that it must be visited and revisited time after time over many months or years before it cracked, hopefully with a big payoff; and other problems much easier, with quicker payoffs. Through it all, Wheeler gave just barely enough advice to keep us from totally floundering, never so much that we felt he had solved our problem for us.

My first problem was a lulu: Take a bar magnet with a magnetic field threading through it and emerging from its two ends. The field consists of field lines, which children are taught to make visible using iron filings on a piece of paper with the magnet below it (Figure 7.1a). Adjacent field lines repel each other. (Their repulsion is felt when one pushes the north poles of two magnets toward each other.) Each magnet's field lines are held together, despite their mutual repulsion, by the magnet's iron. Remove the iron, and